Jakapun Tachaiya

Lecturer · Natural Language Processing researcher

Department of Linguistics, Faculty of Arts, Chulalongkorn University · Bangkok, Thailand

jakapun.t@chula.ac.th
jakapun-t.github.io
github.com/jakapun-t
ORCID 0000-0001-9779-1571

ABOUT

As a faculty member in the Linguistics Department at Chulalongkorn University, I am deeply interested in how computers can understand human language, a complex and often ambiguous task. My research focuses on Natural Language Processing (NLP), enabling computers to perform language-based tasks such as question answering, translation, and large-scale text analysis. Currently, I am developing NLP solutions using Large Language Models (LLMs), particularly for the Thai language. I welcome collaborations in this field.

RESEARCH

Core areas
Large Language Models, Retrieval-Augmented Generation (RAG), Sentiment Analysis, Social Trend and Behavior Analysis, Machine Learning

Applications
Conversational AI, Information Extraction, Text Summarization, Multilingual NLP, Educational Technology, Healthcare NLP

Highlights

- **NLP research:** advancing natural language understanding and generation through novel architectures and methodologies.
- **Collaborations:** working with leading research institutions and industry partners globally.
- **Impact:** research applied in real-world applications affecting millions of users.

EXPERIENCE

2025–present — **Lecturer**
Department of Linguistics, Chulalongkorn University, Thailand

2023–2025 — **Lecturer**
Computer Science Department, KMITL, Thailand

2021–2022 — **Research Lab Leader**
UC Riverside, USA · Led student NLP and data-science projects on social media and contributed to an NSF-funded grant

2016–2017 — **Research Assistant**
Spatial Sciences Institute, USC, USA · Spatial data analysis with NoSQL

EDUCATION

2017–2022 — **Ph.D. in Computer Science**
University of California, Riverside, USA · Advisor: Michalis Faloutsos

2014–2016 — **M.S. in Computer Science**
University of Southern California, USA

2009–2012 — **B.E. in Computer Engineering**
Chulalongkorn University, Thailand

PUBLICATIONS

2024 — **Investigating the Influence of Psychologists' Recommendations on Thai Juvenile Court Judgements**
Praingcharoenkit, C., Sangkhao, S., Kawto, W., Rutherford, A. T., & **Tachaiya, J.** · JCSSE 2024, Bangkok, Thailand · doi:10.1109/jcsse61278.2024.10613718

2022 — **IKEA: Unsupervised Domain-Specific Keyword Expansion**
Tachaiya, J., Gharibshah, J., Imani, A., Papalexakis, E. E., & Faloutsos, M. · ASONAM 2022, Istanbul, Turkey · doi:10.1109/asonam55973.2022.10088556

2021 **SentiStance: Quantifying the Intertwined Changes of Sentiment and Stance in Response to an Event in Online Forums**
Tachaiya, J., Gharibshah, J., Irani, A., Esterling, K. M., & Faloutsos, M. · ASONAM 2021, The Hague, Netherlands · doi:10.1145/3467351.3490966

2020 **RAFFMAN: Measuring and Analyzing Sentiment in Online Political Forum Discussions with an Application to the Trump Impeachment**
Tachaiya, J., Gharibshah, J., & Esterling, K. M. · ICWSM, Atlanta, Georgia · doi:10.1609/icwsm.v15i1.18096

2020 **RThread: A Thread-Centric Analysis of Security Forums**
Tachaiya, J., Gharibshah, J., Papalexakis, E. E., & Faloutsos, M. · ASONAM 2020, The Hague, Netherlands · doi:10.1109/asonam49781.2020.9381312

TEACHING

CHULALONGKORN UNIVERSITY

2025–2026 **Natural Language Processing**
ISE · Undergraduate, international · Fall 2026, Fall 2025

2025–2026 **Statistics for Digital Humanities**
LangTech · Undergraduate, Thai · Fall 2026, Fall 2025

2024–2026 **Programming for Natural Language Processing**
LangTech · Undergraduate, Thai · Fall 2026, Fall 2025, Fall 2024

2025–2026 **Large Language Models**
LangTech · Graduate, Thai · Spring 2026, Spring 2025

KMITL

2022–2024 **Intro to Natural Language Processing**
Undergraduate, Thai · 5 terms

2023–2024 **Foundation of Programming**
Undergraduate, international · Fall 2024, Fall 2023

2023–2024 **Intro to Data Science**
Undergraduate, international · Spring 2024, Spring 2023

2024 **Database Programming in Practice**
Undergraduate, international · Spring 2024

2024 **Generative AI**
Graduate, Thai · Fall 2024

2024 **Data Science Programming & Tools**
Graduate, Thai · Fall 2024

2022 **Web Mining**
Undergraduate, Thai · Fall 2022

TALKS

Jul 2025 **Introduction to LLMs and Generative AI**
Workshop · Thammasat Medical School

Feb 2025 **Generative AI for Daily Work**
Workshop · EC GROUP Head Office

2024–2025 **Generative AI for Daily Work**
Workshop · Thailand Institute of Scientific and Technological Research (TISTR) · April 2024, September 2024, April 2025

Sep 2024 **Accelerating the Machine Learning Process with LLMs: Beyond Traditional Task-Specific Training**
Virtual workshop · KMITL

References and the full publication list are available on request.